

Z': A Proposed Directed Output of the TADA Model

SNP-to-gene pipeline via MAGMA in a common variant GWAS leads to principled pathway analysis.

Multi-marker Analysis of GenoMic Annotation (MAGMA) is a gene-level and gene-set analysis tool where at its core is a multiple regression model defined by

$$Y = \alpha_{0,g} + X_g \alpha_g + W \beta_g + \epsilon_g$$

where Y is a vector of summary Z statistics per SNP, dependent upon: X_g , a matrix of SNP genotypes, α_g , the effect of interest, W , a matrix of covariates, β_g , the coefficients of the covariates, and ϵ_g , the residual error, striated across each gene g . The test for gene-level association assesses whether the joint effect of SNPs across the gene is significantly nonzero,

using an F-test or equivalent. The SNP-wise mean is calculated in which each gene-level p-value is converted to a Z-score via

$$Z_g = \hat{\alpha}_g \cdot \Phi^{-1}(1 - p_g)$$

where Φ^{-1} is the inverse normal (probit) transformation and p_g is the gene-level p-value from the regression. This Z-score can thus be modeled by the full frequentist definition of a Z-statistic:

$$Z = \frac{\hat{\theta} - \theta_0}{SD(\hat{\theta})}$$

The Z-statistic is preferable given its directionality, as dictated by the parameter magnitude in reference to its value under the null. Moreover, the sign of α_g preserves the SNP effects in the calculation of the Z score, preserving the biological relevance of gene enrichment (+ Z) and depletion (-Z). Therefore, using this statistic is ideal for downstream pathway analysis to rank genes of interest by both magnitude and direction. This model, however, works mainly in conjunction with common-variant GWAS data, as rare-variant analyses typically require other statistical methods to compensate for lower power under the same assumptions.

Managing statistical power in rare-variant analysis leaves biological relevance in pathway analysis behind.

In the instance of ASD research, the probabilistic Transmission And De novo Association (TADA) model has been developed using a Bayesian statistical framework to prioritize risk genes weighted by the potential functional severity of their SNPs (e.g. loss-of-function >

missense > synonymous). In adaptations such as TADA+, ‘+’ refers to the incorporation of functional genomic annotations as informative priors, allowing biologically relevant features to up- or down-weight genes before the data are considered. The resulting q-value is a directionless metric bounded by 0 and 1, and the number of unassociated genes skews the q-value distribution toward 1. Relatively data-driven q-value thresholds have been identified to categorize regions of the post-cutoff distribution by strength of association with ASD. For example, the remaining data from a q-threshold of 0.1 represents the retention of ASD-suggestive genes, used in exploratory analysis. While innovative, the output of this model is not as ideal for downstream pathway analysis as the MAGMA Z-statistic given the lack of depletion representation, removing effect as information of interest. Here we seek to employ established statistical theories to extract a Z-like statistic, Z' , from the TADA+ Bayesian model to maintain biological relevance in downstream pathway enrichment analyses and corroborate that gene enrichment across variant classes is statistically tractable.

Deriving Z' .

Given the data are discrete rare-variant counts whose exposure measure is their mutation rate over time, each data point is an independently, but not identically, distributed Poisson random variable X_k :

$$X_k \stackrel{\text{ind.}}{\sim} \text{Poisson}(n \cdot \mu_k \cdot \gamma^{I(\text{case})})$$

where n is the number of mutations, μ_k is the mutation rate per weighted mutation class, and γ is

$$I(\text{case}) = \begin{cases} 1 & \text{if case} \\ 0 & \text{if control} \end{cases}$$

the relative risk parameter determined by the indicator function

Thus we define γ as a multiplicative factor that scales the baseline in cases relative to controls. Under the null, $\gamma = 1$, with no meaningful differentiation in risk between cases and controls.

The likelihood function is derived from the product of these Poisson random variables, where the data vector $\mathbf{x}$ is conditioned on this parameter γ :

$$\mathcal{L}(\mathbf{x} \mid \gamma) = \prod_k \text{Poisson}(x_k \mid n \cdot \mu_k \cdot \gamma^{I(\text{case})})$$

While the likelihood $L(\mathbf{x} \mid \gamma)$ characterizes the probability of the observed mutation counts given the relative risk parameter, our goal is inference on γ itself given the data. We therefore seek the posterior distribution $p(\gamma \mid \mathbf{x})$, which we approximate using the Bernstein-von Mises (BvM) theorem rather than taking advantage of a Gamma-Poisson conjugate prior:¹⁸

$$\sqrt{n}(\gamma - \hat{\gamma}_{\text{MLE}}) \mid \mathbf{x} \xrightarrow{d} \mathcal{N}(0, \mathcal{I}(\gamma_0)^{-1})$$

Here we apply BvM which states that under regular conditions, regardless of distribution, as our sample size n approaches infinity, the prior washes out, and a posterior on γ centered around the MLE is asymptotically Gaussian with the mean being the MLE and a posterior variance converging to the inverse Fisher information (Cramér-Rao lower bound). Standardizing this posterior by subtracting the null value $\gamma=1$ and dividing by the posterior standard deviation yields what we introduce as Z' , converging in distribution to $N(0,1)$ under H_0 :

$$Z' = \frac{\hat{\gamma} - 1}{\text{SD}(\gamma \mid \mathbf{x})}$$

Upon derivation, it was found that remaining on the natural scale rather than log transforming was optimal as $\gamma=1$ is a satisfying biological null, and a log transform would obscure the very directionality we seek to retain. Thus, our Z' result has the following definition of directionality:

$$Z' \begin{cases} > 0 & \hat{\gamma} > 1 & \text{enrichment} \\ = 0 & \hat{\gamma} = 1 & \text{null} \\ < 0 & \hat{\gamma} < 1 & \text{depletion} \end{cases}$$

The main constraints in applying our findings in biological context are sample size, independence (i.e. ancestry), and relative dispersion. The BvM requires a large enough sample size for the posterior to concentrate around the MLE. For our purposes, large rare-variant analyses such as seen in Satterstrom et al. 2020 ($n = 35,584$) are more than sufficient in ensuring this condition holds. Standard QC measures in genetic association analyses such as PCA and proper reference panels satisfy the condition of ensuring SNP independence, along with the assumption that mutations in different classes arise from independent events. Finally, a key characteristic of the Poisson distribution is that its variance equals the expected value, and in turn, the BvM requires consistency in likelihood. In the case of mutation counts being more variable than expected, this can overinflate our Z' . It is also worth noting that the mutation rate μ_k , varying across strata, is not a constant; it itself carries a Gamma prior, making the TADA+

model hierarchical in nature. This does not ultimately change how our Z' is constructed, but this extra term increases how conservative our statistic is.

The resulting Z' -statistic can then be used to rank the genes accordingly and feed them into FGSEA, effectively replacing the q-value. Ultimately, the generation of this Z' -statistic could help evaluate and tighten the aforementioned results, further propagate ASD research in the cross-variant pathway analysis space, and decrease the chance of erroneous assumptions about underlying ASD biology.

CITATIONS

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